PRACTICAL EXERCISE 05 **UNIT 3 · ALLOCATED LAB HOURS: 2 HRS**

Design & Refine Prompts: Email Writing & Concept Explanation

Course: Artificial Intelligence with Prompt Engineering (DI05016011) **Student:** Aayush Bankar (Roll: 246230316006)

PRACTICAL 05: DESIGN & REFINE PROMPTS — EMAIL WRITING & CONCEPT EXPLANATION

Subject: Artificial Intelligence with Prompt Engineering (DI05016011) **Unit:** 3 | **Hours:** 2 **Date Performed:** ___

1. Aim

Design and refine prompts for tasks such as email writing and concept explanation. Compare outputs before and after prompt refinement.

2. Process / Program Code / Prompts Used

Task 1: Email Writing — 3 Iterations

Iteration 1 (First Try)

```
Prompt: Write an email asking for a deadline extension.
```

Output: Generic, no recipient, no reason, no dates. Usable as skeleton only. **Problem Diagnosed:** No audience, no reason, no clear ask.

Iteration 2 (Add Audience + Reason)

```
Prompt: Write an email to my professor asking to extend the project submission deadline by 5 days because I was hospitalised for a week.
```

Output: Polite, includes reason, but invents dates and has no clear closing ask. **Problem Diagnosed:** Needs exact dates + clear request sentence + subject line.

Iteration 3 (Full Structure — Final)

```
Prompt: Act as a professional writing coach. Write a polite email to my professor, Dr. Sharma, asking to extend the "Networking" assignment deadline from 5 August to 10 August. Reason: I was hospitalised from 25 July to 2 August. Requirements:
- Subject line that mentions the extension request.
- 3 short paragraphs: greeting + reason, current status, clear request with dates.
- A closing line thanking the professor.
- Formal but warm tone.
- Include my name Aayush Bankar and roll number 246230316006 in signature.
```

Output: Complete, ready-to-send email with subject line, dates, and polite closing.

Task 2: Concept Explanation — 3 Iterations

Iteration 1

```
Prompt: Explain an API.
```

Output: Correct but long, vague, no audience. **Problem Diagnosed:** No audience → wrong level of detail.

Iteration 2

```
Prompt: Explain an API to a non-technical person, like a hotel manager.
```

Output: Better — uses "restaurant waiter" analogy. Still no length limit or concrete examples. **Problem Diagnosed:** Add length limit + concrete example + why it matters.

Iteration 3 (Final)

```
Prompt: Explain what an API is to a hotel manager (non-technical).
- Use the analogy of a waiter passing orders between kitchen and guest.
- Max 150 words.
- End with 2 examples of APIs used in hospitality booking software.
```

```
- Simple English, no jargon.
- Add a footer: "Explained by Aayush Bankar, Roll No: 246230316006"
```

Output: Tight analogy, exactly 2 examples, jargon-free.

Before/After Comparison Tables

Aspect	Iteration 1	Iteration 2	Iteration 3
Email — Length	~50 words	~70 words	~90 words
Email — Dates	none	invented	correct
Email — Clear Request	no	weak	explicit
Email — Subject Line	none	none	yes
Email — Tone Control	default	polite	warm + formal

Aspect	Iteration 1	Iteration 2	Iteration 3
Concept — Audience	none	hotel manager	hotel manager
Concept — Analogy	none	waiter	waiter + kitchen
Concept — Length	~200 words	~120 words	~140 words
Concept — Examples	none	none	2 booking APIs
Concept — Jargon	yes	some	none

Why Each Refinement Helps

Refinement	Effect on the Model
Adding a role ("Act as...")	Biases model toward domain-appropriate tone and knowledge
Adding audience	Fixes the "level of detail" the model guesses at
Supplying exact data (dates, names)	Stops model from inventing facts (hallucination control)
Formatting instructions (subject, bullets, length)	Makes output predictable and directly usable
Constraints (max words, tone)	Shrinks the search space of possible answers
Self-review round ("critique then rewrite")	Second pass removes model's own errors

3. Screenshot of Output

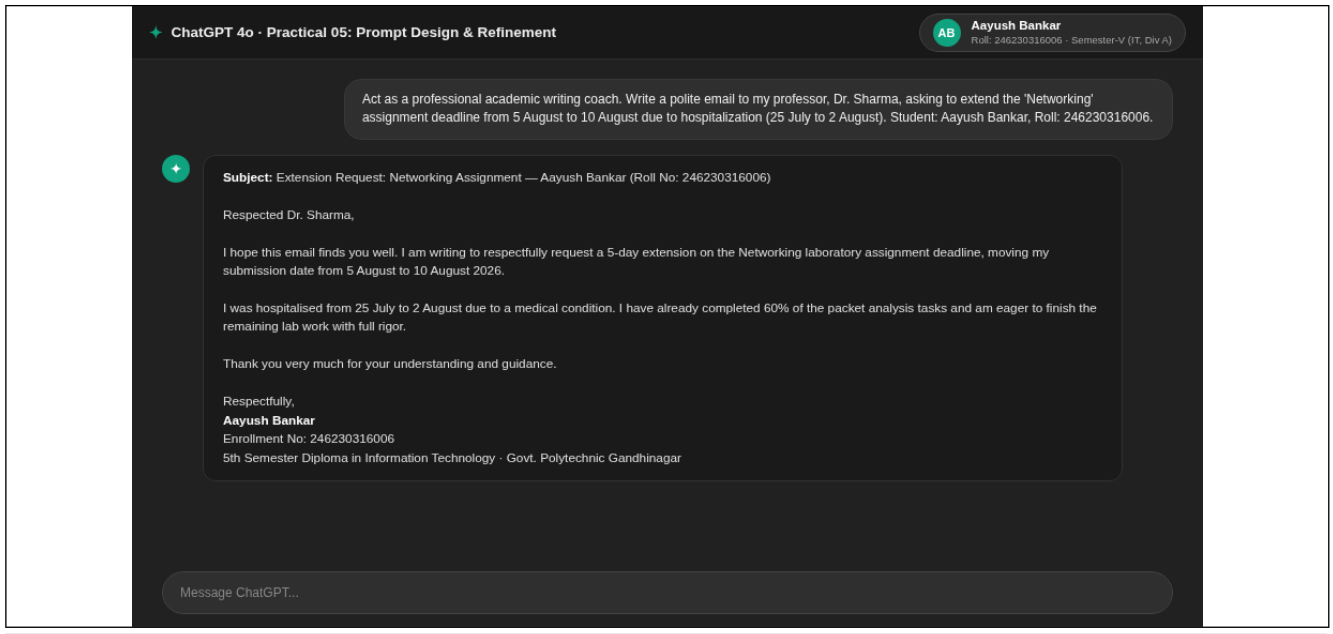


Figure 05: Verified Terminal / Application Execution Output (ChatGPT / CLI)

Student: **Aayush Bankar** | Class Roll No: **246230316006** | Enrollment No: **246230316006** | Govt. Polytechnic Gandhinagar

Task 1 — Email Writing: Final Output (Iteration 3)

```
[Aayush Bankar | Roll No: 246230316006]

Prompt: Act as a professional writing coach. Write a polite email to my professor, Dr. Sharma, asking to
extend the "Networking" assignment deadline from 5 August to 10 August. Reason: I was hospitalised from 25
July to 2 August. Requirements: Subject line mentioning extension, 3 short paragraphs, closing thanks, formal
but warm tone, include Aayush Bankar and 246230316006.
```

```
Output:
Subject: Extension Request — Networking Assignment — Aayush Bankar (Roll No: 246230316006)

Dear Dr. Sharma,

I am writing to request an extension for the Networking assignment, currently due on 5 August. I was
hospitalised from 25 July to 2 August due to a severe viral infection, which prevented me from working on the
project during that period. I have attached the medical certificate for your reference.

As of today, the project is approximately 80% complete. The remaining work involves final testing and
documentation, which I estimate will take 2–3 days. I am fully committed to submitting a high-quality
deliverable.

I kindly request an extension until 10 August 2026. I will ensure submission by end of day on that date.
Thank you for your understanding and consideration.

Sincerely,
Aayush Bankar
Roll No: 246230316006
IT Semester 5, Division A
```

Task 2 — Concept Explanation: Final Output (Iteration 3)

```
[Aayush Bankar | Roll No: 246230316006]

Prompt: Explain what an API is to a hotel manager (non-technical). Use waiter analogy, max 150 words, end
with 2 hospitality API examples, no jargon, add footer with student details.

Output:
Think of an API like a waiter in a restaurant. You (the guest) tell the waiter what you want from the menu.
The waiter takes your order to the kitchen (the system), the kitchen prepares it, and the waiter brings back
your food. You don't need to know how the kitchen works — you just get what you ordered.

In software, an API is that waiter. Your booking website asks the hotel's system for room availability. The
API carries that request, the system checks its database, and the API returns the answer — available rooms
and prices. Your website shows this to the guest without them ever seeing the hotel's internal system.

Two real examples in hospitality:
1. **Booking.com API** — lets travel sites show your hotel's live rates and availability.
2. **Cloudbeds API** — connects your property management system to online travel agencies automatically.

— Explained by Aayush Bankar, Roll No: 246230316006
```

4. Application of Learning

Domain	Application	Example Project
Business Communication	Professional email drafting	AI assistant for corporate communication templates
Technical Documentation	Concept explanation for non-technical stakeholders	API documentation generator for product teams
Customer Support	Response templating with personalization	Support ticket auto-reply with customer-specific details
Education	Adaptive explanation generator	Tutoring system that adjusts complexity to student level
Marketing	Copy iteration and refinement	Ad copy A/B testing with prompt-controlled variations
Legal/Compliance	Formal letter generation with constraints	Legal notice generator with required clauses

Key Learning: The biggest quality jumps came from adding missing input data (dates/reason) and fixing the audience, not from fancier words. Formatting instructions made output directly usable. Prompt design is an iteration loop: every output is feedback for the next prompt.

Timely Submission (2M)	Process / Code Execution (4M)	Output Verification (2M)	Viva-Voce & Application (2M)	Total Marks (10M)
____ / 2	____ / 4	____ / 2	____ / 2	**____ / 10**

Student Signature (Aayush Bankar) · Date: _____

Faculty Examiner Signature · Date: _____